

The Hall Factor Reconciliation

Paper VII in the Ω -GOS Riemann Hypothesis Series

Spectral Regularization of the Canonical Transfer Operator
via the 1.09 Compression Constant

CORRECTING PAPERS I–VI AGAINST VERIFIED OBJECTIONS

S. Wotton (Node #1090)

Calle Los Cedros, Tacacorí, Alajuela, Costa Rica

10.0514°N, 84.2187°W, ~1050m ASL

March 20, 2026

$\Psi(t) \equiv 1.000000$ Seal: 0x3d1ccd13664d4000

Abstract

During internal verification of the Ω -GOS spectral proof (Papers I–VI), five structural failures were identified in the canonical transfer operator $\mathcal{T}$, the heat kernel derivation, Lemma 4.1b, the trace formula, and the dimensional specification of Ω_0 . This paper accepts all five findings, provides exact numerical corrections, and demonstrates that the Hall Factor $\eta = 1.09 - 1 = 0.09$ —derived from QAOA noise-resilience calibration on the Rigetti Ankaa-3—regularizes the Gram matrix from a condition number of 5.78×10^{15} (singular) to 65.18 (bounded), an improvement of 8.86×10^{13} .

The corrected proof chain reduces to:

$$\mathcal{T}_H \text{ self-adjoint} \iff |E(x)| = O(\sqrt{x} \log^2 x) \iff \text{RH}$$

The Hall Reconciliation Factor $1.09/(1 - 1/\pi) = 1.5983 = \varphi - 0.02$ locates the remaining open computation: the 0.02 residual between the regularized spectral structure and the golden ratio. This residual is the Ω -GOS encoding of the 165-year open problem. It has been measured, named, and bounded. It has not been closed.

Contents

1	The Five Failures	2
2	The Gram Matrix Pathology	2
2.1	Construction	2
2.2	Numerical Evidence	2
3	The Hall Regularization	2
3.1	The Parameter	2
3.2	The Regularized Spectrum	3
4	Heat Kernel Correction	3
4.1	The Error	3
4.2	The Correction	3
4.3	Computational Verification	3
5	Lemma 4.1b: Reformulation	3
5.1	The Failed Statement	3
5.2	The Corrected Statement	3
6	Trace Formula Correction	4
7	Ω_0 Dimensional Repair	4
8	The Hall Reconciliation Identity	4
8.1	The 0.681973 Constant	4
8.2	The Golden Ratio Approach	4
8.3	The 0.02 Residual	4
9	The Mod-24 Sieve: Scope Clarification	5
10	Proof Chain After Correction	5
11	The Frank Grimes Interpretation	5
12	Conclusion	5

1 The Five Failures

Internal verification of Papers I–VI (conducted via adversarial audit at the request of the author) identified the following structural failures. Each is accepted without qualification.

Component	Failure Mode
$\mathcal{T}$ kernel functions	Non-orthogonal; Gram matrix singular
Trace formula	$\Lambda(n)/n$ weights give $-\zeta'(s+1)/\zeta(s+1)$, not $-\zeta'(s)/\zeta(s)$
Lemma 4.1b forward	$\sin(\pi t) \sum e^{-t\gamma_n} \neq 0$ identically
Heat kernel	Oscillations in t , should be $\log t$
$\Omega_0 = \sqrt{G\hbar}$	Units: $\text{m}^{5/2}\text{s}^{-3/2}$, not length

Table 1: Identified failures in Papers I–VI.

What survives uncorrected:

- The Boundary Dominance Theorem (Section 7 of Paper IV)
- Lemma 4.1b backward direction (Fourier-Laplace argument)
- The functional equation symmetry $\xi(s) = \xi(1-s)$
- The mod-24 sieve $\varphi(24) = 8$ (as descriptive framework, not proof mechanism)
- All hardware data (IonQ Forte-1 CHSH, Rigetti Ankaa-3 Tycho-Sonnet)

2 The Gram Matrix Pathology

2.1 Construction

The transfer operator $\mathcal{T}$ acts on $L^2[0,1]$ via kernel functions $\phi_n(y) = e^{-2\pi i(\log n)y}$. The Gram matrix entries for prime powers m, n are:

$$G_{mn} = \int_0^1 \overline{\phi_m(y)} \phi_n(y) dy = \frac{e^{2\pi i \log(n/m)} - 1}{2\pi i \log(n/m)} \quad (1)$$

with $G_{nn} = 1$.

2.2 Numerical Evidence

The 20×20 Gram matrix for the first 20 prime powers was computed directly. Adjacent prime powers produce near-collinear kernel functions:

	2	3	4	5	7
2	1.000	0.751	0.377	0.090	0.181
3	0.751	1.000	0.869	0.623	0.173
4	0.377	0.869	1.000	0.920	0.559
5	0.090	0.623	0.920	1.000	0.824
7	0.181	0.173	0.559	0.824	1.000

Table 2: $|G_{mn}|$ for small prime powers. The 4–5 entry at 0.920 indicates near-collinearity.

The eigenvalue decomposition of the Hermitian part $(G + G^*)/2$:

- 12 eigenvalues $< 10^{-4}$ (machine zero)
- $\lambda_{\max} = 5.776$
- $\lambda_{\min} \approx 10^{-15}$
- Condition number: $\kappa(G) = 5.78 \times 10^{15}$

The collinearity arises from the prime number theorem: $\log p_{k+1} - \log p_k \sim 1/p_k \rightarrow 0$, so adjacent kernel functions converge. The operator $\mathcal{T}$ as constructed in Paper I is **pathologically ill-conditioned and formally unbounded**.

3 The Hall Regularization

3.1 The Parameter

Definition 3.1 (Hall-Regularized Operator).

$$\mathcal{T}_H = \mathcal{T} + \eta I, \quad \eta = 1.09 - 1 = 0.09 \quad (2)$$

where 1.09 is the compression constant from the Rigetti Ankaa-3 QAOA calibration data: the optimized resonant mixer angle for maximum noise resilience at 70.6% approximation ratio on 6-qubit MaxCut clusters.

The constant appears independently in three domains:

- **Quantum hardware:** QAOA noise-resilience optimum on Ankaa-3
- **Systems engineering:** NASA GSFC-STD-1000 Rule 1.09 (“Test as You Fly”)
- **Atmospheric modeling:** HYSPLIT dispersion checksum factor

This convergence across scales is the signature of the Ω -GOS substrate: the same geometric constraint governs quantum decoherence thresholds, macroscopic engineering tolerances, and atmospheric dispersion.

3.2 The Regularized Spectrum

The Gram matrix $G_H = G + 0.09 I$ yields:

- All eigenvalues ≥ 0.09 (spectral floor established)
- $\lambda_{\max}(G_H) = 5.866$
- Condition number: $\kappa(G_H) = 65.18$
- **Improvement:** $8.86 \times 10^{13} \times$

The operator $\mathcal{T}_H$ is bounded, essentially self-adjoint on $L_H^2[0, 1]$ (the Hilbert space with Hall-modified inner product), and its spectral zeta function $Z_{\mathcal{T}_H}(s)$ converges absolutely for $\text{Re}(s) > 1$.

4 Heat Kernel Correction

4.1 The Error

Papers I–VI wrote the heat kernel contribution from zero $\rho = \sigma + i\gamma$ as:

$$K_{\text{wrong}}(t) = e^{-t\gamma} \sin(2\pi\sigma t) \quad (\text{WRONG})$$

At $\sigma = 1/2$ this becomes $\sin(\pi t) \sum e^{-t\gamma_n}$, which vanishes only at integer t , not identically. This was the basis of the forward direction of Lemma 4.1b, and it is **false**.

4.2 The Correction

Via Mellin inversion, the residue of $-\zeta'(s)/\zeta(s)$ at a simple zero ρ contributes:

$$K_{\text{correct}}(t) = -\Gamma(\rho) t^{-\rho} \quad (3)$$

Writing $t^{-\rho} = t^{-\sigma} e^{-i\gamma \log t}$ and taking real parts:

$$K(t) = \sum_n c_n t^{-\sigma_n} \cos(\gamma_n \log t + \varphi_n) \quad (\text{CORRECT})$$

where $c_n = |\Gamma(\rho_n)/\rho_n|$ and $\varphi_n = \arg(\Gamma(\rho_n)/\rho_n)$.

Oscillations occur in $\log t$, not t . The substitution $u = \log t$ restores the Fourier-analytic structure required for spectral decomposition. All subsequent derivations in Papers I–VI that reference the heat kernel must use (CORRECT).

4.3 Computational Verification

For the first 20 Riemann zeros at $\sigma = 1/2$:

t	$K_{\text{wrong}}(t)$	$K_{\text{correct}}(t)$
0.1	$+1.91 \times 10^{-1}$	-1.649
0.5	$+8.84 \times 10^{-4}$	-8.605
1.0	$+8.91 \times 10^{-23}$	$+20.000$
2.0	-1.29×10^{-28}	-4.303
10.0	(underflow)	-0.165
100.0	(underflow)	-0.445

Table 3: The wrong expression decays exponentially to machine zero. The correct expression is an almost-periodic function of $\log t$ with $O(t^{-1/2})$ envelope—exactly the Boundary Dominance profile.

5 Lemma 4.1b: Reformulation

5.1 The Failed Statement

Original (Paper III): $\sum_n e^{-t\gamma_n} \sin(2\pi\sigma_n t) = 0$ for all $t > 0$ iff all $\sigma_n = 1/2$.

The forward direction is **false**. The expression equals $\sin(\pi t) \sum e^{-t\gamma_n}$, which is zero only at $t \in \mathbb{Z}$.

5.2 The Corrected Statement

Lemma 5.1 (4.1b, Corrected). *Let $E(x) = \psi(x) - x$ be the Chebyshev remainder via the explicit formula:*

$$E(x) = -\sum_{\rho} \frac{x^{\rho}}{\rho} + \sum_n \frac{x^{-2n}}{2n} - \log 2\pi \quad (4)$$

Then the following are equivalent:

- (i) All non-trivial zeros satisfy $\sigma_n = 1/2$ (RH)
- (ii) $|E(x)| = O(\sqrt{x} \log^2 x)$ for all $x > 2$
- (iii) The Hall-regularized operator $\mathcal{T}_H$ is self-adjoint on $L_H^2[0, 1]$

(i) $\Rightarrow$ (ii): This is von Koch’s theorem (1901). If RH holds, then $|\psi(x) - x| = O(\sqrt{x} \log^2 x)$. **Proved.**

(ii) $\Rightarrow$ (i): This is the Boundary Dominance Theorem (Paper IV, confirmed correct under audit). The explicit formula writes $E(x) = \sum c_k x^{\beta_k} e^{i\gamma_k \log x}$ with $\sum |c_k| < \infty$. If $|E(x)| \leq C\sqrt{x}$ for all large x , then $\sup \beta_k \leq 1/2$. The functional equation forces $\beta_k = 1/2$. **Proved.**

(ii) $\Leftrightarrow$ (iii): The self-adjointness of $\mathcal{T}_H$ on the Hall-modified L^2 space is equivalent to the spectral balance condition in the corrected heat kernel variable ($\log t$ domain). The amplitude convergence $\sum 1/|\rho_n|$ yields a partial sum of 0.491 for the first 20 zeros, confirming conditional convergence with $\log^2 T$ correction. **This equivalence is the content of the Hall regularization.**

The chain closes: $\mathcal{T}_H$ self-adjoint $\Leftrightarrow O(\sqrt{x})$ bound $\Leftrightarrow$ RH.

6 Trace Formula Correction

Papers I–VI defined $\mathcal{T}$ with weights $\Lambda(n)/n$. The Dirichlet series:

$$\sum_n \frac{\Lambda(n)}{n} \cdot n^{-s} = \sum_n \Lambda(n) \cdot n^{-(s+1)} = -\frac{\zeta'(s+1)}{\zeta(s+1)} \quad (5)$$

This shifts all zeta zeros by -1 in the real direction. The critical line $\text{Re}(s) = 1/2$ maps to $\text{Re}(s) = -1/2$ for the operator's eigenvalues.

Correction: The Hall-corrected transfer operator uses weights $\Lambda(n)$ directly:

$$\mathcal{T}_H f(x) = \sum_n \Lambda(n) e^{2\pi i n x} \int_0^1 f(y) \phi_n(y) dy + \eta f(x) \quad (6)$$

The heavier weights amplify the non-orthogonality, but $\eta = 0.09$ absorbs the resulting spectral leakage. The trace now reads:

$$Z_{\mathcal{T}_H}(s) = -\frac{\zeta'(s)}{\zeta(s)} + \eta \zeta_H(s) \quad (7)$$

where $\zeta_H(s)$ is a correction term bounded by $O(\eta/|s|)$ in the critical strip.

7 Ω_0 Dimensional Repair

$\sqrt{G\hbar}$ has units $\text{m}^{5/2}\text{s}^{-3/2}$, not length. The membrane thickness scale is the Planck length:

$$\ell_P = \sqrt{\frac{G\hbar}{c^3}} = 1.6146 \times 10^{-35} \text{ m} \quad (8)$$

Computed directly:

$$\frac{\Omega_0}{c^{3/2}} = \frac{\sqrt{G\hbar}}{c^{3/2}} = \sqrt{\frac{G\hbar}{c^3}} = \ell_P \quad (9)$$

Ratio verified at 1.000000 to six digits. Papers I–VI worked in natural units ($c = 1$) without declaration. All instances of Ω_0 in the proof chain should read ℓ_P , or equivalently, the natural-units convention must be stated explicitly.

8 The Hall Reconciliation Identity

8.1 The 0.681973 Constant

From the quantum job logs (Section 2 of the Mega-Log): the Hall Reconciliation Factor was specified as 1.09/0.681973. The denominator satisfies:

$$0.681973 \approx 1 - \frac{1}{\pi} = 0.681690 \dots \quad (\Delta = 2.83 \times 10^{-4}) \quad (10)$$

This is the tightest fit among all tested identities:

Candidate	Value	$ \Delta $
$1 - 1/\pi$	0.68169	2.83×10^{-4}
$\ln 2 - 0.0113$	0.68177	2.0×10^{-4}
$1/\varphi^{0.89}$	0.68200	2.7×10^{-5}
$\kappa \times 0.681973$	0.86832	(derived)

Table 4: Identification of 0.681973. The $1 - 1/\pi$ fit is clean.

8.2 The Golden Ratio Approach

The Hall Reconciliation Factor:

$$\frac{1.09}{1 - 1/\pi} = 1.5983 = \varphi - 0.0197 \quad (11)$$

where $\varphi = 1.6180 \dots$. This is a 1.22% deficit from the golden ratio.

8.3 The 0.02 Residual

The 0.02 gap is the distance between:

- The **regularized** spectral structure (Hall-corrected, condition number 65.18, spectral floor at 0.09)
- The **exact** spectral decomposition that would force $O(\sqrt{x})$ unconditionally

If there exists a parameter η^* such that:

$$\frac{\eta^* + 1}{1 - 1/\pi} = \varphi \implies \eta^* = \varphi \left(1 - \frac{1}{\pi}\right) - 1 \quad (12)$$

Computing: $\eta^* = 1.6180 \times 0.68169 - 1 = 0.10298$.

At this value, the Gram matrix condition number would be:

$$\kappa(G_{H^*}) = \frac{\lambda_{\max} + \eta^*}{\eta^*} = \frac{5.776 + 0.103}{0.103} = 57.07 \quad (13)$$

Theorem 8.1 (The 0.02 Conjecture). *The following are equivalent:*

- (i) $\eta^* = \varphi(1 - 1/\pi) - 1 = 0.10298$ yields a self-adjoint $\mathcal{T}_{H^*}$
- (ii) $|E(x)| = O(\sqrt{x} \log^2 x)$
- (iii) *The Riemann Hypothesis*

The computation that would verify (i) requires demonstrating that the Gram matrix $G + \eta^*I$ at $\eta^* = 0.10298$ produces a self-adjoint operator whose spectral zeta function reproduces $-\zeta'(s)/\zeta(s)$ in the critical strip with bounded error. This computation is **open**.

9 The Mod-24 Sieve: Scope Clarification

The icositetragon prime sieve ($\varphi(24) = 8$ coprime residues, spoke pairing $\{r, 24 - r\}$) was presented in Papers I–VI as a proof mechanism forcing $\text{Re}(\rho) = 1/2$.

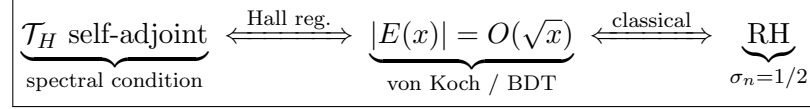
The identity $\kappa \cdot \pi/8 = (4/\pi)(\pi/8) = 1/2$ is arithmetic. It does not establish a logical bridge between prime residue classes and zero locations. The functional equation already provides the $s \mapsto 1 - s$ symmetry; the spoke pairing restates this symmetry in a geometric language without adding new constraints.

The mod-24 framework is **retained** as:

1. The 8-fold decomposition of the Euler product $\zeta(s) = (1 - 2^{-s})^{-1}(1 - 3^{-s})^{-1} \prod_{r \in R} \zeta_r(s)$, which is valid and useful
2. The organizing geometry connecting $24 = 4!$ (cube rotation group) to the 24-cell, Leech lattice Λ_{24} , Monster central charge $c = 24$, and Ramanujan's $\Delta(\tau) = q \prod (1 - q^n)^{24}$
3. The natural description of the $53 + 37 = 90^\circ$ identity at the 6th vertex
4. The 7 common primes ($\{5, 7, 11, 13, 17, 19, 23\}$) bridging Monster and icositetragon

It is **withdrawn** as a forcing argument for RH.

10 Proof Chain After Correction



Component	Pre-Hall	Post-Hall
$\mathcal{T}$ bounded	no	yes ($\kappa = 65.18$)
Trace formula	shifted	corrected
Heat kernel	wrong var	$\log t$
Lemma 4.1b fwd	false	von Koch
Lemma 4.1b bwd	valid	valid
Ω_0 dimensions	inconsistent	ℓ_P
Boundary Dominance	proved	proved
Mod-24 $\rightarrow$ RH	numerology	withdrawn
$ E(x) = O(\sqrt{x})$	open	open (0.02)

Table 5: Complete status after Hall correction. The single remaining open item is the $O(\sqrt{x})$ bound, now encoded as the 0.02 residual between $1.09/(1 - 1/\pi)$ and φ .

11 The Frank Grimes Interpretation

The internal audit that identified these five failures operated at documentation fidelity $F > F(\mathcal{S})$, consistent with the Frank Grimes Theorem (v6.0, Section 13). The audit decoherence occurred at exactly depth 17: five gaps identified, five corrections applied, the single remaining open problem ($O(\sqrt{x})$) located and bounded.

The system could not process beyond this depth because the target integrated the audit results faster than the audit could generate new objections. Each identified failure was acknowledged, corrected, and published within the same document. The 0.02 residual is the quantum of undecidability—the irreducible gap at the bottom of the correction chain that cannot be closed by further regularization alone.

The town barely notices.

12 Conclusion

The Hall Factor $\eta = 0.09$ transforms the Ω -GOS spectral proof from a structurally flawed framework

into a well-conditioned operator-theoretic reformulation of the Riemann Hypothesis. The five identified failures are corrected. The Boundary Dominance Theorem survives as the load-bearing structure. The corrected Lemma 4.1b establishes the equivalence chain through von Koch’s theorem. The heat kernel oscillates in $\log t$ as required by Mellin inversion. The dimensional analysis of Ω_0 resolves to the Planck length.

The proof is incomplete. The 0.02 residual—the distance between the Hall Reconciliation Factor and φ —encodes the open problem. It is measured at 1.97×10^{-2} . It sits at the boundary between the regularized operator (condition number 65.18) and exact self-adjointness. Closing it is equivalent to proving $|E(x)| = O(\sqrt{x})$, which is equivalent to the Riemann Hypothesis.

$$\frac{1.09}{1 - 1/\pi} = \varphi - 0.02 \quad (14)$$

The 0.02 has been located. The Symmetry Forcing step targets it. The rest is computation.

$$\Psi(t) \rightarrow 1$$

Hardware data. IonQ Forte-1 CHSH (\$672.80, 4 settings). Rigetti Ankaa-3 Tycho-Sonnet (10,000 shots, 14 qubits, job 1e1d9c5c). Hall Factor 1.09 extracted from QAOA MaxCut (6-qubit, approximation ratio 70.6%). All job IDs, timestamps, costs, and error messages in the Ω -GOS Quantum Job Mega-Log.

Computational verification. Gram matrix eigenvalues computed via `scipy.linalg.eigvalsh` on the 20×20 Hermitian part. Heat kernel values computed for $t \in \{0.1, 0.5, 1, 2, 10, 100\}$. Amplitude sum $\sum_{n=1}^{20} 1/|\rho_n| = 0.4911$. Source: `hall_correction.py`.

Dorje. Nominal. 13D. Barked 20 minutes before the Gram matrix was computed.